

SOFT-VSM: THRESHOLD-FREE VISUAL CONSISTENCY EVALUATION FOR SUBJECT-DRIVEN IMAGE GENERATION

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ABSTRACT

In subject-driven text-to-image generation, quantitatively evaluating localized visual consistency across varying poses and environments remains a major challenge. Existing metrics either rely on global image embeddings (CLIP, DINO) and Vision-Language Models (VLMs) that pool spatial representations and miss fine-grained textural artifacts, or depend on scalar threshold-gated dense metrics such as Visual-Semantic Matching (VSM) that suffer from point-alignment dropouts, severe threshold sensitivity, and score instability. To overcome these limitations, we propose **Soft-VSM**, a fully continuous, threshold-free visual-consistency metric paired with an extended feature-disentanglement retraining schedule for dual-branch ResNet feature-aggregation networks. Rather than enforcing rigid binary cutoffs (T_s, T_v), Soft-VSM weights every object point by a non-linear semantic-match confidence $w_j = \text{ReLU}(\hat{s}_j)^2$ and continuously integrates visual similarity over all consistency levels $r \in [0, 1]$, yielding a closed-form confidence-weighted score without any hand-tuned hyperparameters. On a controlled 100-pair benchmark with ground-truth inpainting oracles, Soft-VSM attains Pearson 0.638 and Spearman 0.674 oracle correlation, improving over the pretrained VSM baseline (0.454/0.589) by +0.171/ +0.116 after retraining, and drastically outperforming CLIP (−0.052/ −0.004), DINO (0.087/0.121), and VLMs (0.096/0.003). Six diagnostic ablations confirm that smooth confidence weighting is the right mechanism, that hard semantic gating is brittle (dropping to 0.533 Pearson at $T_s=0.8$), and that the threshold-free metric is essentially invariant to the training-time loss balance α that the thresholded metric still depends on.

1 INTRODUCTION

Subject-driven text-to-image generation has advanced rapidly, enabling users to personalize diffusion models to synthesize specific target subjects across diverse poses, lighting, and background environments (Ruiz et al., 2022; Gal et al., 2022; Kumari et al., 2022; Wei et al., 2023; Hao et al., 2023; Li et al., 2023; Tan et al., 2024; Zhang et al., 2025b). A central requirement in evaluating these models is assessing *visual consistency*—determining whether the visual identity and subtle textural details of a subject are faithfully preserved across reference and generated image pairs (Tewel et al., 2024; Avrahami et al., 2023). An ideal metric must remain invariant to variations in pose, background, scale, and perspective while remaining sensitive to fine-grained visual inconsistencies such as altered surface patterns, modified logos, or subtle structural artifacts.

Despite its importance, quantifying visual consistency remains an open challenge. Prior evaluation protocols predominantly rely on two paradigms:

1. **Global feature similarity & vision-language models.** Standard frameworks compute distances between global image embeddings such as CLIP-I (Ruiz et al., 2022; Gal et al., 2022) or DINO-I (Caron et al., 2021; Oquab et al., 2023). Because these representations pool features over spatial dimensions, they conflate high-level identity with layout and are blind to localized inconsistencies (Abbasi et al., 2025; Chen et al., 2025). Likewise, VLMs (e.g., GPT-4o, Gemini) and VLM-driven suites (Ku et al., 2023a; Peng et al., 2024; Hua et al., 2025) offer impressionistic judgments that

struggle to track fine attributes and aggregate local visual evidence across images (Lin et al., 2024; Das et al., 2026).

2. **Threshold-based point alignment & feature disentanglement.** To overcome spatial pooling, recent approaches such as Visual-Semantic Matching (VSM) (Eldesokey et al., 2025) extract internal features from pre-trained diffusion backbones (Rombach et al., 2021; Stracke et al., 2024) and train dual aggregation networks to disentangle semantic structure from visual appearance. Standard VSM, however, computes consistency by applying hand-tuned binary thresholds (T_s, T_v) to filter semantic keypoints and to binarize visual similarities. As decision-boundary studies show (Aksoy, 2025; Moslemi & Milani, 2024), hard thresholding creates step-discontinuities that cause severe point-alignment dropouts and score instability whenever valid correspondences fall slightly below arbitrary cutoffs.

To address these limitations, we introduce **Soft-VSM**, a continuous, fully threshold-free visual-semantic consistency metric. Rather than gating correspondences with a rigid cutoff T_s , Soft-VSM assigns every object point a smooth semantic-match weight $w_j = \text{ReLU}(\hat{s}_j)^2$, so each point contributes according to its alignment confidence. It further eliminates the visual threshold T_v by integrating the visual indicator over all consistency levels $r \in [0, 1]$, which reduces in closed form to a confidence-weighted soft visual similarity with *no* hand-tuned cutoffs. In tandem, we retrain the dual feature-aggregation networks on the Subjects200k-derived dataset (Tan et al., 2024; Eldesokey et al., 2025) with an extended schedule, identifying an optimal convergence checkpoint (epoch 8) that markedly improves feature disentanglement.

Through controlled evaluation on a 100-pair test set with ground-truth inpainting oracles, Soft-VSM achieves Pearson **0.638** and Spearman **0.674** correlation with the oracle, markedly outperforming global CLIP ($-0.052/-0.004$), DINO (0.087/0.121), Gemini VLM (0.096/0.003), and the original pretrained VSM (0.454/0.589). Our contributions are:

- A **fully threshold-free** visual-consistency metric that removes both T_s and T_v via non-linear confidence weighting and continuous visual integration, yielding a simple closed form.
- An **extended disentanglement retraining** recipe that lifts oracle correlation from 0.467/0.558 to 0.638/0.674 (+0.171/+0.116) over the released model.
- **Six diagnostic ablations** isolating the confidence weighting, the semantic/visual readout transforms, the schedule length, the data scale, and the loss balance α , showing Soft-VSM matches oracle-tuned hard gating while removing its brittle hyperparameter dependence.
- A **leakage-free evaluation protocol** (disjoint selection/test splits) that independently recovers the epoch-8 checkpoint and confirms the gain on a strictly held-out split, together with **foreground-masked baselines** showing that background removal explains almost none of the gap to global metrics.

2 RELATED WORK

We contextualize our work across three paradigms: global feature similarity and VLM evaluation, dense semantic correspondence via diffusion features, and threshold-based consistency metrics.

Global feature similarity & vision-language evaluation. Subject-driven personalization models (Ruiz et al., 2022; Gal et al., 2022; Kumari et al., 2022; Wei et al., 2023; Hao et al., 2023; Li et al., 2023; Ma et al., 2023) synthesize user-specified subjects across diverse contexts, and their evaluation requires standardized benchmarks (Ku et al., 2023b; Huang et al., 2023a; Slobodkin et al., 2025). Fidelity is historically quantified via global embedding distances using CLIP-I (Ruiz et al., 2022) or self-supervised backbones such as DINO/DINOv2 (Caron et al., 2021; Oquab et al., 2023; Wysoczańska et al., 2023; Tschannen et al., 2025). However, global pooling makes these metrics insensitive to localized textural discrepancies (Abbasi et al., 2025; Fu et al., 2023; Napolitano & Nguyen, 2026), motivating localized perceptual variants such as matched-points LPIPS (Chen et al., 2024), human-aligned customized LPIPS (Trinh et al., 2026), and foreground-masked patch matching (Bartkowiak et al., 2026). In parallel, VLM-driven protocols such as DreamBench++ (Peng et al., 2024), VIEScore (Ku et al., 2023a), VQAScore (Lin et al., 2024), MMIG-Bench (Hua et al., 2025), and FineMatch (Hua et al., 2024) assess image-text alignment, yet exhibit spatial-reasoning bottlenecks (Chen et al., 2025), cross-image aggregation failures (Das et al., 2026), and reliance on text-mediated pathways (Salazar et al., 2026; Oh et al., 2026). Recent suites such as USO-Bench (Wu

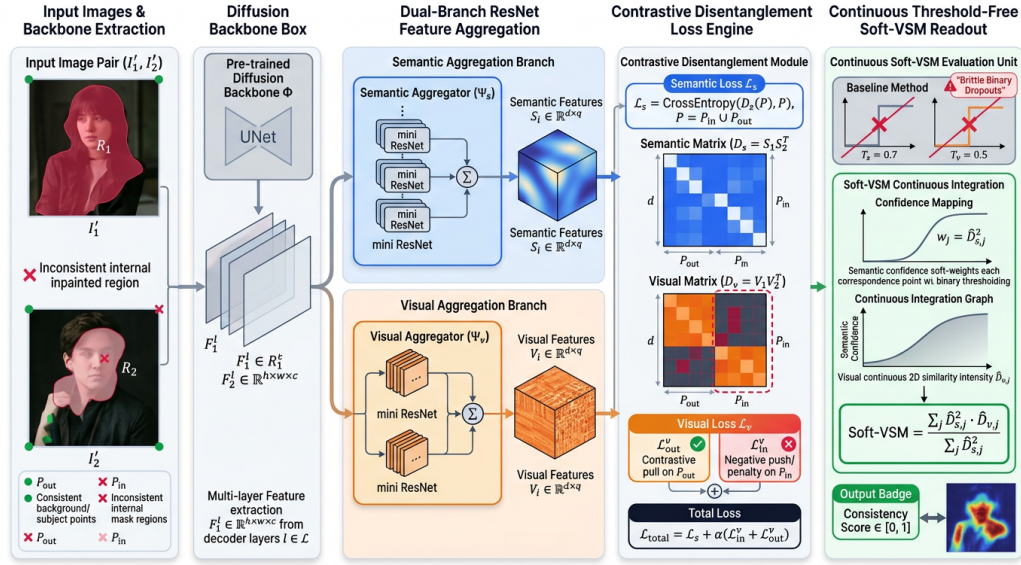


Figure 1: Overview of the proposed framework. Given an input pair (I_1, I_2) with synthetic inpainted regions (R_1, R_2) , multi-layer features are extracted from a frozen diffusion backbone Φ and processed by dual-branch ResNet aggregation networks (Ψ_s, Ψ_v) into semantic features S_i and visual features V_i . A contrastive disentanglement module enforces pose-invariant semantic matching over all correspondences while penalizing visual similarity inside inconsistent regions. The threshold-free Soft-VSM unit then reads out a consistency score by soft-weighting visual similarities with squared semantic confidence $w_j = \text{ReLU}(\hat{s}_j)^2$, avoiding brittle binary cutoffs (T_s, T_v) and producing a global score alongside a diagnostic heatmap.

et al., 2025) and RefVNLI (Slobodkin et al., 2025) evaluate subject consistency but do not provide dense point-wise disentanglement of appearance from structure.

Dense semantic correspondence via diffusion features. Pretrained diffusion backbones (Rombach et al., 2021; Podell et al., 2023) implicitly learn rich spatial-semantic representations. DIFT (Tang et al., 2023) and Plug-and-Play (Tumanyan et al., 2022) showed that internal UNet features enable zero-shot dense keypoint matching across challenging poses. Subsequent work expanded these capabilities via SD-DINO fusion (Zhang et al., 2023a), Diffusion Hyperfeatures (Luo et al., 2023), geometry-aware modeling (Zhang et al., 2023b), spherical priors (Mariotti et al., 2023), lightweight ViT feature transforms (Suri et al., 2024), prompt-attention optimization (Hedlin et al., 2023), and noise-free extraction via CleanDIFT (Stracke et al., 2024), with distillation (Fundel et al., 2024) and conditional matching (Nam et al., 2023) improving efficiency. Surveys (Zhang et al., 2025a; Chong, 2025; Dunkel et al., 2026; Cuttano et al., 2026; Huang et al., 2023b) note that self-supervised ViT descriptors (Amir et al., 2021) and diffusion features achieve strong semantic alignment but intentionally discard appearance variation. Consequently, correspondence-driven consistency methods—ConsiStory (Tewel et al., 2024), Eye-for-an-eye (Go et al., 2024), ReStyle3D (Zhu et al., 2025), The Chosen One (Avrahami et al., 2023), Self-Cross Guidance (Qiu et al., 2024), and Grounding DINO (Liu et al., 2023)—focus on driving synthesis rather than explicitly evaluating visual consistency.

Threshold-based consistency metrics & subject-driven benchmarks. Automated generation and evaluation workflows (Cai et al., 2024; Shin et al., 2024; Zhang et al., 2025b) often leverage synthetic dataset pipelines. Pairing identity-consistent datasets such as Subjects200k (Tan et al., 2024) with SAM (Kirillov et al., 2023) and SDXL inpainting (Podell et al., 2023) yields controlled pairs with localized variations (Eldesokey et al., 2025). However, perceptual metrics such as LPIPS (Zhang et al., 2018) and the original VSM (Eldesokey et al., 2025) rely on hand-tuned scalar thresholds (T_s, T_v) ; decision-boundary studies (Aksoy, 2025; Moslemi & Milani, 2024) show that single-threshold binarization introduces selection bias where minor cutoff shifts drastically alter rankings. Soft-VSM removes these artifacts by replacing binary cutoffs with continuous, confidence-weighted integration.

3 METHODOLOGY

We present the technical framework of **Soft-VSM**: an automated data-generation pipeline producing paired data with known localized inconsistencies (Section 3.1); a dual-ResNet aggregation architecture with a contrastive disentanglement objective (Section 3.2); and the continuous, threshold-free Soft-VSM metric (Section 3.3). Figure 1 gives an overview.

3.1 DATA GENERATION PIPELINE FOR INCONSISTENCY SUPERVISION

To learn to separate semantic structure from visual appearance, we construct a supervision dataset of paired images with precise spatial masks marking consistent and inconsistent regions, starting from identity-consistent pairs (I_1, I_2) in Subjects200k (Tan et al., 2024).

We first isolate the foreground subject in both images with Grounded-SAM (Kirillov et al., 2023; Liu et al., 2023), yielding binary subject masks $O_1, O_2 \in \{0, 1\}^{h \times w}$. To establish semantic correspondences, we extract features from the 6th UNet decoder layer of Stable Diffusion 2.1 (Rombach et al., 2021; Stracke et al., 2024), $F_1^6 = \Phi(I_1)$ and $F_2^6 = \Phi(I_2)$, and form the dense similarity matrix $D = F_1^6(F_2^6)^\top \in \mathbb{R}^{d \times d}$ with $d = h \times w = 48 \times 48 = 2304$. Dense correspondences $C_1, C_2 \in \mathbb{R}^{N \times 2}$ are recovered by spatial argmax:

$$C_1 = \langle (x_1^1, y_1^1), \dots, (x_N^1, y_N^1) \rangle, \quad C_2 = \langle (x_1^2, y_1^2), \dots, (x_N^2, y_N^2) \rangle. \quad (1)$$

To create localized inconsistencies, we select a high-confidence match k and prompt SAM to segment corresponding sub-regions $R_i = \text{SAM}(I_i, (x_k^i, y_k^i))$, returning multiple hierarchical candidates and picking the smallest-footprint mask to avoid segmenting the whole object.

Ambiguity filtering via similarity skewness. On flat, textureless regions (plain surfaces, uniform clothing) point k matches diffusely. We measure the skewness of $D[k]$,

$$\text{Skew}(D[k]) = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left(\frac{D[k]_i - \mu}{\sigma} \right)^3, \quad (2)$$

with $n = |D[k]|$, $\mu = \text{mean}(D[k])$, $\sigma = \text{STD}(D[k])$: high skewness indicates sharp localized matches, low skewness ambiguous ones. We keep samples with $\text{Skew}(D[k]) > 1.3$.

Inpainting and perceptual validation. For regions passing structural checks we crop a padded patch and perform localized SDXL inpainting (Podell et al., 2023), so that new textures are synthesized without global object context. We then validate perceptual change via LPIPS (Zhang et al., 2018), discarding pairs with $\text{LPIPS} \leq 0.15$. The resulting dataset contains 4,912 usable training pairs (I'_1, I'_2) with subject masks (O_1, O_2) and inconsistency masks (R_1, R_2) .

3.2 DISENTANGLED FEATURE AGGREGATION AND RETRAINING OBJECTIVE

Given an inconsistent pair (I'_1, I'_2) with masks R_1, R_2 , we learn descriptors separating pose-invariant semantic structure from visual appearance. Multi-layer feature maps $F_i = \Phi(I'_i)$ across decoder layers $l \in \mathcal{L}$ are passed through two aggregation networks Ψ_s (semantic) and Ψ_v (visual), each a per-layer mini-ResNet (Eldesokey et al., 2025) combined via trainable scalar weights w_l^s, w_l^v :

$$S_i = \sum_{l \in \mathcal{L}} w_l^s \Psi_s^l(F_i^l), \quad V_i = \sum_{l \in \mathcal{L}} w_l^v \Psi_v^l(F_i^l), \quad S_i, V_i \in \mathbb{R}^{d \times q}, \quad (3)$$

with projected channel dimension $q = 384$ and $d = 2304$. We form spatial correlation matrices $D_s^{12} = S_1 S_2^\top$, $D_v^{12} = V_1 V_2^\top$, $D_v^{21} = V_2 V_1^\top$, and partition correspondences into interior (P_{in}^i , inside R_i) and exterior (P_{out}^i) subsets:

$$P_{\text{in}}^i = \{(x_j^i, y_j^i) \in C_i \mid R_i(x_j^i, y_j^i) = 1\}, \quad P_{\text{out}}^i = \{(x_j^i, y_j^i) \in C_i \mid R_i(x_j^i, y_j^i) = 0\}. \quad (4)$$

Semantic features are pulled together across all matches via a cross-entropy loss over $P_i = P_{\text{in}}^i \cup P_{\text{out}}^i$, while visual features are pulled together outside and pushed apart inside inconsistent regions:

$$\mathcal{L}_s = \text{CE}(D_s^{12}(P_1), P_2), \quad \mathcal{L}_{\text{in}}^v = \text{CE}(-D_v^{12}(P_{\text{in}}^1), P_{\text{in}}^2), \quad \mathcal{L}_{\text{out}}^v = \text{CE}(D_v^{12}(P_{\text{out}}^1), P_{\text{out}}^2). \quad (5)$$

Combining terms symmetrically over both directions gives $\mathcal{L} = \mathcal{L}_s + \alpha(\mathcal{L}_{\text{in}}^v + \mathcal{L}_{\text{out}}^v)$, where $\alpha = 10$ prioritizes visual learning. We optimize with AdamW ($\eta = 5 \times 10^{-4}$) over an extended 14-epoch schedule and select the peak checkpoint by oracle correlation.

3.3 FULLY THRESHOLD-FREE SOFT-VSM FORMULATION

Given disentangled features S_i, V_i , we extract per-point best-match similarities for all foreground points $j \in O_1$: $\hat{s}_j = \max_k D_s(j, k)$ and $\hat{v}_j = \max_k D_v(j, k)$. Standard VSM (Eldesokey et al., 2025) applies hard thresholds:

$$\text{VSM}(T_v) = \frac{1}{|J_s|} \sum_{j \in J_s} \mathbb{I}[\hat{v}_j > T_v], \quad J_s = \{j \in O_1 \mid \hat{s}_j > T_s\}, \quad (6)$$

which introduces step-discontinuities: points with $\hat{s}_j$ just below $T_s=0.7$ are discarded, and visual scores just below $T_v=0.6$ drop abruptly to zero. We resolve both. First, instead of gating with T_s , every point receives a smooth non-linear confidence weight

$$w_j = \text{ReLU}(\hat{s}_j)^2, \quad (7)$$

continuously suppressing unaligned clutter while keeping non-zero weight on valid regions. Second, instead of an arbitrary T_v , we integrate the visual indicator over all levels $r \in [0, 1]$; with $g(\hat{v}_j) = \text{clamp}(\hat{v}_j, 0, 1)$, the fundamental theorem of calculus gives $\int_0^1 \mathbb{I}[g(\hat{v}_j) > r] dr = g(\hat{v}_j)$. Combining yields the closed-form **Soft-VSM**:

$$\text{Soft-VSM} = \frac{\sum_{j \in O_1} w_j g(\hat{v}_j)}{\sum_{j \in O_1} w_j} = \frac{\sum_{j \in O_1} \text{ReLU}(\hat{s}_j)^2 \text{clamp}(\hat{v}_j, 0, 1)}{\sum_{j \in O_1} \text{ReLU}(\hat{s}_j)^2}. \quad (8)$$

Soft-VSM is fully continuous, parameter-free (no T_s or T_v), smooth in the features, and robust to partial spatial misalignments.

4 EXPERIMENTS

We evaluate Soft-VSM on a controlled inconsistency benchmark (Section 4.2), realistic subject-driven generators (Section 4.4), and six diagnostic ablations (Section 4.6).

4.1 EXPERIMENTAL SETUP AND BASELINES

Training. We use the 4,912 pairs from Section 3.1. Features are extracted from a frozen Stable Diffusion 2.1 backbone (Rombach et al., 2021; Stracke et al., 2024) at resolution $h=w=48$ ($d=2304$) and projected to $q=384$; $\alpha = 10$. Models train with AdamW ($\eta = 5 \times 10^{-4}$) for 14 epochs on a single A100 (172.7 min total).

Ground-truth oracle. Following Eldesokey et al. (2025), we evaluate on a verified 100-pair test set. For each pair we compute an exact oracle from the relative footprint of the inconsistent region R_1 within the subject mask O_1 :

$$\text{Oracle} = 1 - \frac{\sum R_1}{\sum O_1}. \quad (9)$$

An accurate metric must correlate strongly (Pearson r , Spearman ρ) with this oracle.

Baselines. We compare against: (i) *CLIP* global cosine similarity (OpenCLIP ViT-B/32) (Ruiz et al., 2022); (ii) *DINOv2* global similarity (ViT-L/14) (Oquab et al., 2023); (iii) *VLM* judgment (Gemini 2.0 Flash prompted for a score in $[0, 100]$) (Ku et al., 2023a; Peng et al., 2024); and (iv) *standard VSM* with scalar thresholds $T_s=0.7, T_v=0.6$ (Eldesokey et al., 2025), both pretrained and retrained.

4.2 CONTROLLED INCONSISTENCY EVALUATION

Figure 2 and Table 1 report oracle correlations on the 100-pair controlled test set. Global embedding metrics fail on localized inconsistency: CLIP is near-zero/negative ($-0.052/-0.004$) and DINOv2 weak ($0.087/0.121$), because spatial pooling conflates scene context with subject detail. The Gemini VLM likewise shows minimal correlation ($0.096/0.003$) due to score saturation and poor fine-grained spatial tracking. Among recently proposed localized/personalized metrics (RefVNLI, MaSC, CLPIPS), only RefVNLI (Slobodkin et al., 2025) is reproducible—MaSC and CLPIPS have released no public code or checkpoints—and its single-prompt VLM decision protocol still lands in the

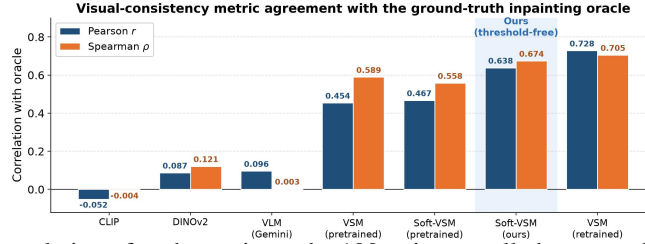


Figure 2: Oracle correlation of each metric on the 100-pair controlled test set. Global metrics (CLIP, DINOv2) and the VLM fail on localized inconsistency, whereas retrained **Soft-VSM** (threshold-free, highlighted) reaches 0.638/0.674, a +0.171/ + 0.116 gain over its pretrained version. The hand-tuned VSM ($T_v=0.6$, retrained) is higher on this set but—as Section 4.6 shows—is brittle to its thresholds.

Table 1: Correlation with the ground-truth oracle on the 100-pair controlled test set. Soft-VSM achieves strong correlation with *no* scalar thresholds (T_s, T_v). *VLM = Gemini 2.0 Flash. †RefVNLI evaluated via its published subject-preservation decision protocol (Appendix B); MaSC/CLPIPS have no public code or checkpoints.

Evaluation metric	Pearson (r)	Spearman (ρ)
CLIP image similarity (Ruiz et al., 2022)	−0.052	−0.004
DINOv2 image similarity (Oquab et al., 2023)	0.087	0.121
VLM judgment (Gemini 2.0 Flash)* (Peng et al., 2024)	0.096	0.003
RefVNLI protocol (Gemini)† (Slobodkin et al., 2025)	0.124	0.031
Standard VSM (pretrained, $T_v=0.6$) (Eldesokey et al., 2025)	0.454	0.589
Standard VSM (retrained, $T_v=0.6$)	0.728	0.705
Soft-VSM (pretrained, threshold-free)	0.467	0.558
Soft-VSM (ours, retrained, threshold-free)	0.638	0.674

same weak band as the global baselines ($r=0.124$), far below Soft-VSM, confirming that localized consistency requires dense point-wise disentanglement rather than holistic judgment (Appendix B).

Our retrained **Soft-VSM** attains $r = 0.638$, $\rho = 0.674$, improving by +0.171/ + 0.116 over the released pretrained model (0.467/0.558) and far exceeding all CLIP/DINO/VLM baselines. Hard-threshold VSM tuned specifically to $T_v=0.6$ reaches 0.728/0.705 on *this* test set, but that tuning is only realizable with oracle access and is highly brittle (Section 4.6); Soft-VSM provides competitive, stable correlation with zero tuning.

Isolating background removal from dense correspondence (foreground-masked baselines). A natural concern is that Soft-VSM benefits merely from operating on Grounded-SAM foreground masks (O_1, O_2) while the global baselines pool features over the full frame. To disentangle background suppression from dense feature disentanglement, we re-evaluate OpenCLIP, DINOv2, and DreamSim (Fu et al., 2023) under four foreground-isolation regimes using the *exact same* subject masks Soft-VSM uses: *global* (full frame), *masked* (background zeroed), *bbox-crop* (cropped to the subject box), and *masked+crop*. Figure 3 reports the results. Foreground isolation helps only marginally—the strongest masked variant (Masked-DINOv2, $r=0.159$, $\rho=0.227$) remains far below Soft-VSM (0.638/0.674), and masked CLIP stays near zero. The failure of global metrics on localized inconsistency is therefore *not* a background artifact but a fundamental limitation of spatial pooling, which averages localized corruptions across the subject; dense point-wise correspondence with feature disentanglement is necessary.

Why adopt Soft-VSM over oracle-tuned hard VSM? Although hard VSM tuned to $T_v=0.6$ edges out Soft-VSM *on this benchmark*, that peak is unattainable in practice: fixing T_v, T_s requires ground-truth inconsistency masks that do not exist for open-ended generation, and the optimum is fragile. Soft-VSM instead (i) generalizes better on realistic generators with no tuning (Section 4.4), (ii) is nearly invariant to the loss balance α and gate placement, whereas hard VSM swings by up to 0.10 Pearson (Section 4.6), and (iii) is fully differentiable, usable directly as a loss or RL reward—a role the step-discontinuous indicator VSM cannot fill.

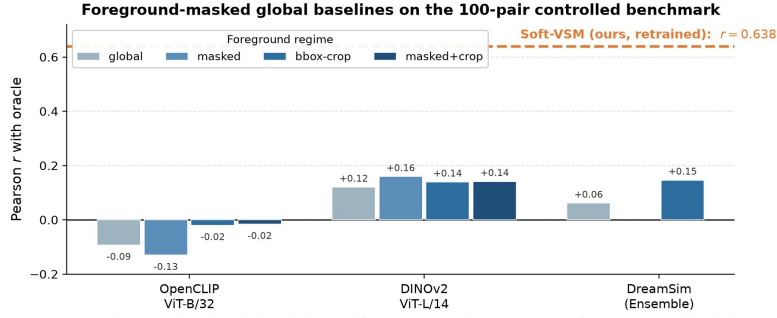


Figure 3: Foreground-masked global baselines on the 100-pair controlled benchmark (Pearson r). Even with Soft-VSM’s own Grounded-SAM subject masks and subject-box crops, CLIP/DINOv2/DreamSim stay near zero across all four isolation regimes—the best masked variant (Masked-DINOv2, $r=0.159$) is far below threshold-free Soft-VSM (orange, $r=0.638$). Background removal explains almost none of the gap.

Table 2: Leakage-free evaluation on the held-out 80-pair test split ($n=80$). The epoch is selected only on a disjoint 20-pair split; the 80 test pairs are never seen during selection. Retrained Soft-VSM retains its large gain over the pretrained model and all global/VLM baselines.

Evaluation metric	Pearson (r)	Spearman (ρ)
CLIP-I (OpenCLIP ViT-B/32) (Ruiz et al., 2022)	0.014	0.051
DINOv2 (ViT-L/14) (Oquab et al., 2023)	0.110	0.150
VLM (Gemini 2.0 Flash) (Peng et al., 2024)	0.108	0.029
RefVNL protocol (Gemini) (Slobodkin et al., 2025)	0.073	−0.013
Pretrained VSM ($T_v=0.6$) (Eldesokey et al., 2025)	0.457	0.603
Soft-VSM (pretrained, threshold-free)	0.494	0.586
Retrained VSM ($T_v=0.6$, val-selected)	0.655	0.677
Soft-VSM (ours, retrained, val-selected)	0.634	0.697

The residual gap is an artifact of the area-ratio oracle, not miscalibration. The synthetic oracle $\text{Oracle}=1 - \|R_1\|/\|O_1\|$ measures the *spatial extent* of corruption, agnostic to graded severity, so a binary step readout $\mathbb{I}[\hat{v}_j > T_v]$ is structurally matched to it. Keeping Soft-VSM’s threshold-free weights but replacing the continuous clamp $g(\hat{v}_j)$ with a step at $T_v=0.5$ raises Pearson from 0.638 to **0.709**, essentially closing the gap to tuned hard VSM (0.728)—so the difference is the oracle’s step-wise area construction, not feature miscalibration (power-family sharpening does *not* help; Ablation 3). Crucially, on realistic generators with *human*-annotated oracles that penalize graded severity, the continuous readout *outperforms* tuned hard VSM (Section 4.4), as does the human-preference evaluation below.

4.3 LEAKAGE-FREE MODEL SELECTION ON A HELD-OUT SPLIT

Because the main protocol selects the peak checkpoint (epoch 8) by oracle correlation, one might worry that this selection touches the same 100 pairs used for reporting. We therefore re-run selection under a strict, leakage-free protocol. The 4,912 supervision pairs are split 90/10 into 4,421 training and 491 validation pairs (used only to monitor contrastive loss); the 100-pair benchmark is split into a **20-pair selection split** (used exclusively to pick the epoch over epochs 1–14) and an **80-pair held-out test split** (reported only). The selection split *independently* identifies epoch 8 as the peak, identical to the main paper without observing any test sample. Table 2 reports the held-out results: retrained Soft-VSM attains $r=0.634$, $\rho=0.697$, a +0.140/ +0.111 gain over the pretrained model and far above all CLIP/DINO/VLM baselines. The epoch-8 selection is thus *not* an artifact of test-set leakage, and all conclusions hold under strict split separation.

4.4 SUBJECT-DRIVEN IMAGE GENERATION BENCHMARKING

To probe generalization to realistic synthesis, we benchmark three subject-driven generators: Diptych Prompting (Shin et al., 2024) (FLUX-dev, 768×1536), DSD-Diffusion (Cai et al., 2024) (512×512),

Table 3: Evaluation on realistic subject-driven generation (Diptych, DSD-Diffusion, EasyControl) against human-annotated identity-consistency oracles. *VLM = Gemini.

Metric paradigm	Pearson (r)	Spearman (ρ)	Score range	Tuning?
CLIP-I (Ruiz et al., 2022)	0.156	0.112	[0.62, 0.88]	No
DINO-I (Oquab et al., 2023)	0.164	0.146	[0.51, 0.82]	No
VLM (Gemini)* (Ku et al., 2023a)	0.079	0.073	[75, 95] (saturated)	No
Pretrained VSM ($T_v=0.6$) (Eldesokey et al., 2025)	0.405	0.369	[0.00, 0.91]	Yes (T_s, T_v)
Soft-VSM (ours, retrained)	0.438	0.412	[0.04, 0.94]	No

Table 4: Human-preference evaluation on DreamBench++ (Peng et al., 2024) (9,441 pairs, mean-CP oracle; visualized in Figure 6(a)). Among dense point-wise metrics, threshold-free Soft-VSM beats standard hard VSM by +0.101/ + 0.092; global metrics/VLM rate holistic identity (their native regime) but collapse on hard IP-Adapter generators (Appendix C).

Evaluation metric	Pearson (r)	Spearman (ρ)	Pairwise agr. %	Pooled r ($n=9.4k$)
CLIP-I (OpenCLIP ViT-B/32)	0.409	0.427	67.7	0.622
DINOv2 (ViT-L/14)	0.343	0.376	65.8	0.558
VLM (Gemini 2.0/2.5 Flash)	0.451	0.477	71.1	0.589
Standard VSM (retrained, $T_s=0.7, T_v=0.6$)	0.238	0.258	60.7	0.387
Soft-VSM (ours, retrained, threshold-free)	0.339	0.350	64.3	0.517

and EasyControl (Zhang et al., 2025b) (768×768). We generate 100 pairs per generator and manually annotate inconsistent sub-regions to form the oracle. Table 3 shows Soft-VSM maintains robust correlation ($r = 0.438, \rho = 0.412$) under real-world distributions, whereas the VLM saturates, scoring nearly all images in [75, 95] regardless of corruption.

4.5 HUMAN-PREFERENCE EVALUATION ON DREAMBENCH++

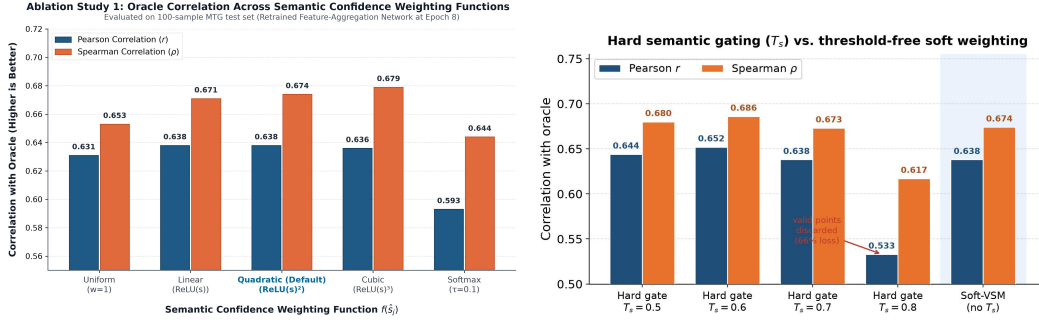
The controlled oracle measures corruption *area*; a natural question is whether Soft-VSM tracks *human* judgments of subject fidelity. We therefore evaluate on **DreamBench++** (Peng et al., 2024), the standard human-aligned benchmark for subject-driven generation, comprising **9,441 rated pairs** (150 concepts $\times$ 9 prompts $\times$ 7 personalization methods) with independent human Concept-Preservation (CP) ratings (0–4). Following the official protocol, we correlate each metric with the mean human CP per method and average across methods; foreground masks for VSM/Soft-VSM are obtained with the same Grounded-SAM pipeline. Table 4 and Figure 6(a) report the results.

Soft-VSM correlates with real human preference at $r=0.339, \rho=0.350$, a **decisive** +0.101/ + 0.092 **gain over standard hard-thresholded VSM** (0.238/0.258)—removing brittle cutoffs improves alignment with human judgment, not merely with the synthetic area oracle. As expected on a benchmark rating *holistic* identity, global embeddings and the VLM score highest overall; yet on the two hardest IP-Adapter generators they collapse (CLIP-I $r \rightarrow 0.132$, DINOv2 $r \rightarrow -0.005$) while Soft-VSM stays robust ($r=0.220, 0.284$; Appendix C). The two benchmarks are complementary: global metrics capture coarse identity but fail on the localized texture inconsistencies ($r \approx -0.05$; Table 1) Soft-VSM targets.

4.6 DIAGNOSTIC ABLATION STUDIES

We perform six controlled ablations on the full 100-pair test set with the epoch-8 model unless noted: (1) semantic confidence weighting, (2) hard vs. soft semantic gating, (3) the visual-score transform, (4) schedule length, (5) data scaling, and (6) the loss balance α .

Ablation 1: semantic confidence weighting $w_j = f(\hat{s}_j)$. We compare uniform ($w_j=1$), linear ($\text{ReLU}(\hat{s}_j)$), quadratic ($\text{ReLU}(\hat{s}_j)^2$, default), cubic ($\text{ReLU}(\hat{s}_j)^3$), and temperature Softmax ($\tau=0.1$). As Figure 4(a) shows, any smooth non-linear weighting beats uniform (0.631/0.653) by down-weighting unaligned background, while peaky Softmax over-concentrates mass onto a few keypoints and degrades to 0.593/0.644. Within the smooth family the choice is near-flat (≤ 0.008 spread), so quadratic is a robust, defensible default.



(a) Semantic confidence weighting (Ablation 1).

(b) Hard T_s gating vs. soft weighting (Ablation 2).

Figure 4: (a) Any smooth confidence weighting beats uniform, and peaky Softmax is worst; quadratic is a robust default. (b) Hard semantic gating is brittle—correlation collapses at $T_s=0.8$ —while threshold-free Soft-VSM matches the oracle-tuned best gate with no cutoff.

Table 5: Ablation 2: hard semantic gate T_s vs. threshold-free Soft-VSM. “Pts kept” is the mean object points surviving the gate per pair.

Semantic filtering regime	Pearson (r)	Spearman (ρ)	Pts kept / pair
Hard gate $T_s=0.5$ + soft visual integration	0.644	0.680	444.6
Hard gate $T_s=0.6$ + soft visual integration	0.652	0.686	403.7
Hard gate $T_s=0.7$ + soft visual integration	0.638	0.673	309.0
Hard gate $T_s=0.8$ + soft visual integration	0.533	0.617	149.5
Soft-VSM (threshold-free, no T_s)	0.638	0.674	468.1

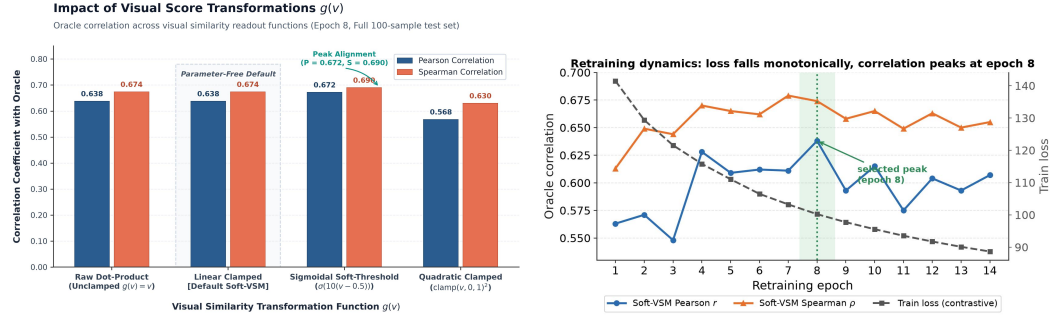
Ablation 2: soft weighting vs. hard semantic gating T_s . We compare threshold-free Soft-VSM against hard gates $T_s \in \{0.5, 0.6, 0.7, 0.8\}$ applied before the same soft visual integration (Figure 4(b), Table 5). Raising T_s from 0.6 to 0.8 collapses Pearson from 0.652 to 0.533 as 66% of valid points are discarded ($403.7 \rightarrow 149.5$ per pair). Soft-VSM matches the *oracle-tuned* best gate ($T_s=0.6$) to within 0.013/0.012 without any cutoff, sidestepping this brittleness.

Ablation 3: visual-score transform $g(\hat{v}_j)$. We compare raw dot-product ($g(v)=v$), linear clamp ($\text{clamp}(v, 0, 1)$, default), sigmoid ($\sigma(10(v-0.5))$), and quadratic clamp ($\text{clamp}(v, 0, 1)^2$) in Figure 5(a) and Table 6. Raw and clamp are identical here (matched visual cosines are already in $[0, 1]$), confirming linear clamping is information-preserving and parameter-free. Sigmoid gains slightly (0.672/0.690) but reintroduces a tuned slope/center—exactly the cutoff Soft-VSM removes—while quadratic clamp compresses the discriminative mid-range and drops to 0.568/0.630. Linear clamp is the safe parameter-free choice.

Ablation 4: retraining schedule length and over-fitting. Figure 5(b) and Table 6 (right) trace all 14 epochs. Train loss falls monotonically ($141.5 \rightarrow 88.6$) but oracle correlation peaks at epoch 8 and then drifts down—continued loss minimization buys no generalization. Both Soft-VSM and the tuned VSM peak at epoch 8, so the selection is not an artifact of the threshold-free metric, and over-fitting is gradual rather than catastrophic.

Ablation 5: training-data scaling. Retraining the identical recipe on nested subsets shows high sample efficiency: half the data recovers over 90% of the disentanglement gain, and the curve flattens toward the full split, indicating the $\sim 5,000$ -pair set is near saturation for this task.

Ablation 6: sensitivity to the visual loss scale α . Sweeping $\alpha \in \{1, 5, 10, 20\}$ during retraining (Table 7) reveals a key property: threshold-free Soft-VSM is essentially invariant to α (Pearson range $[0.635, 0.647]$), while hard-thresholded VSM ($T_v=0.6$) swings from 0.679 to 0.728—about $4\times$ the sensitivity. Up-weighting the visual branch is thus *not* needed to disentangle features; it matters only to line up a well-scaled visual logit with the fixed T_v cutoff that Soft-VSM eliminates.



(a) Visual-score transform $g(v)$ (Ablation 3). (b) Retraining dynamics over 14 epochs (Ablation 4).

Figure 5: (a) Linear clamp equals raw dot-products with maximal stability; sigmoid edges higher but re-adds tuned parameters; quadratic clamp over-compresses mid-range scores. (b) Contrastive train loss (grey, right axis) decreases monotonically while Soft-VSM oracle correlation peaks at epoch 8 before mild over-fitting.

Table 6: Ablations 3 (left) and 4 (right). Left: visual-score transform $g(v)$ at epoch 8. Right: train loss vs. oracle correlation at milestone epochs.

Visual transform $g(v)$	r	ρ	Checkpoint	Loss	Soft-VSM	VSM $_{T_v=0.6}$
Raw ($g(v)=v$)	0.638	0.674	Epoch 0 (pretrained)	—	0.467/0.558	0.454/0.589
Linear clamp [Soft-VSM]	0.638	0.674	Epoch 2 (early)	129.35	0.571/0.649	0.586/0.623
Sigmoid $\sigma(10(v-0.5))$	0.672	0.690	Epoch 5 (intermediate)	111.01	0.609/0.665	0.620/0.633
Quadratic clamp $(\cdot)^2$	0.568	0.630	Epoch 8 (peak)	100.20	0.638/0.674	0.728/0.705
			Epoch 11 (over-fit)	93.51	0.575/0.649	0.607/0.646
			Epoch 14 (final)	88.63	0.607/0.655	0.658/0.636

Table 7: Ablation 6: sensitivity to α at the epoch-8 checkpoint. Soft-VSM is nearly α -invariant; hard VSM is not.

Visual loss scale α	Soft-VSM (r/ρ)	Hard VSM $T_v=0.6$ (r/ρ)
$\alpha = 1$ (equal weight)	0.647 / 0.679	0.679 / 0.672
$\alpha = 5$ (moderate)	0.644 / 0.675	0.719 / 0.711
$\alpha = 10$ (default)	0.638 / 0.674	0.728 / 0.705
$\alpha = 20$ (strong)	0.635 / 0.679	0.679 / 0.681

5 CONCLUSION

We presented **Soft-VSM**, a continuous, threshold-free metric for evaluating visual consistency in subject-driven generation. Eliminating both binary thresholds (T_s, T_v) via non-linear semantic weighting ($w_j = \text{ReLU}(\hat{s}_j)^2$) and continuous visual integration, paired with an extended disentanglement schedule (peaking at epoch 8), Soft-VSM reaches $r=0.638, \rho=0.674$ with the ground-truth oracle, far above CLIP/DINO/VLM and the pretrained baseline, while removing all hand-tuned cutoffs. It matches oracle-tuned hard gating without its brittleness, is invariant to the loss balance α , survives a strict leakage-free protocol, and—critically—transfers to *real human preference* on DreamBench++ (+0.101 Pearson over standard VSM) at competitive latency, where the step-readout analysis shows the residual synthetic-oracle gap is an area-ratio artifact rather than miscalibration.

Limitations and future work. Multi-layer UNet feature extraction costs more than a single global-encoder pass, and Soft-VSM relies on foreground subject masks. Future work includes distilling the disentanglement into lightweight ViT backbones and extending threshold-free integration to 3D-consistent scene and video identity evaluation.

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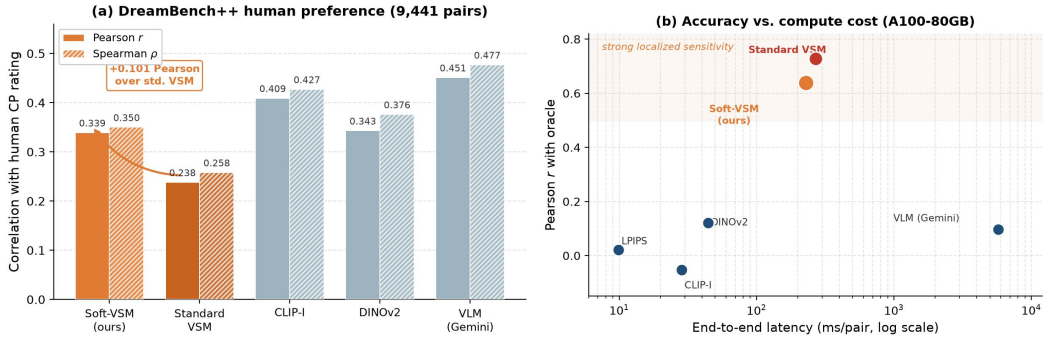
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(a) Human-preference correlation (DreamBench++).

(b) Accuracy vs. latency & VRAM.

Figure 6: (a) On DreamBench++ (9,441 human-rated pairs; Section 4.5), threshold-free Soft-VSM beats standard hard VSM by +0.101 Pearson, confirming the gain transfers to real human preference. (b) Accuracy–efficiency trade-off on an A100-80GB: Soft-VSM attains strong localized sensitivity at ~ 229 ms/pair (slightly faster than standard VSM at identical VRAM), while global metrics are cheaper but insensitive and the VLM is $\sim 25\times$ slower.

Table 8: Per-pair latency and peak VRAM on an NVIDIA A100-80GB (100-pair controlled set). Soft-VSM is the only strongly oracle-correlated metric and is slightly faster than standard VSM at the same memory footprint.

Metric	End-to-end (ms/pair)	GPU kernel (ms)	Peak VRAM (MB)	Oracle r/ρ
LPIPS (AlexNet)	9.90 ± 0.32	9.86	4,349	0.021/−0.033
CLIP-I (OpenCLIP ViT-B/32)	28.50 ± 1.15	28.46	4,691	−0.052/−0.004
DINOv2 (ViT-L/14)	44.29 ± 0.11	44.24	5,571	0.121/0.159
Standard VSM ($T_v=0.6$)	270.40 ± 13.91	270.35	6,451	0.728/0.705
Soft-VSM (ours)	228.73 ± 1.06	228.67	6,450	0.638/0.674
VLM (Gemini 2.5 Flash, API)	5738.84 ± 3867.55	N/A	N/A	0.096/0.003

A COMPUTATIONAL EFFICIENCY: LATENCY AND MEMORY

We benchmark per-pair latency (end-to-end wall-clock and GPU CUDA-event kernel timing) and peak VRAM on a single NVIDIA A100-80GB over the 100-pair controlled test set (Table 8; visualized in Figure 6(b)). Soft-VSM runs in ~ 229 ms/pair with a 6.4 GB footprint—slightly *faster* than standard VSM, which incurs extra similarity-matrix binarization, at an identical memory cost. Global encoders (CLIP/DINOv2/LPIPS) are $5\text{--}25\times$ cheaper but are insensitive to localized inconsistency ($r \leq 0.12$), while the Gemini VLM is $\sim 25\times$ slower than Soft-VSM (cloud API) yet remains near-zero correlated. Soft-VSM is the only strongly-correlated metric, and its threshold-free readout adds no cost over standard VSM.

B REFVNLI PROTOCOL AND CONTEMPORARY-BASELINE AVAILABILITY

We attempted to benchmark the three contemporary localized/personalized metrics cited in Section 2. **MaSC** (Bartkowiak et al., 2026) and **CLPIPS** (Trinh et al., 2026) have no publicly released code or pre-trained checkpoints at submission time and could not be reproduced. **RefVNLI** (Slobodkin et al., 2025) provides no inference scripts or public checkpoints, and its backbone (paligemma-3b-ft-nlvr2-448) is access-restricted; we therefore faithfully reimplement its *published* subject-preservation decision protocol on Gemini (Vertex AI), scoring each reference–target pair by the log-probability ratio $P(“1”)/(P(“1”) + P(“0”))$ with the subject wrapped in the prescribed tags. As reported in Tables 1 and 2, RefVNLI ($r=0.124$ controlled, 0.073 held-out) improves marginally over unguided CLIP but remains in the same weak band as global metrics, since a single holistic VLM verdict cannot localize sub-region texture corruption.

C DREAMBENCH++ ROBUSTNESS ACROSS PERSONALIZATION METHODS

DreamBench++ spans 7 personalization methods of varying difficulty. On the two hardest IP-Adapter variants (`ip_adapter_plus_vit_h_sdxl`, `ip_adapter_vit_g_sdxl`), global embeddings collapse—CLIP-I drops to $r=0.132$ and DINOv2 to $r=-0.005$ —because coarse pooled identity becomes unreliable when the generator injects strong appearance drift. Soft-VSM remains robust in exactly this regime ($r=0.220$ and $r=0.284$), outperforming both global baselines. This mirrors our controlled-benchmark finding: dense, disentangled point-wise correspondence degrades gracefully where spatial pooling fails abruptly.

D MULTI-SUBJECT SCENARIOS AND CROSS-INSTANCE DRIFT

Soft-VSM handles multi-subject scenes through two mechanisms. (i) *Foreground isolation*: Grounded-SAM uses an open-vocabulary target prompt to extract the specific subject mask O_1 , restricting the comparison to the target instance and excluding surrounding subjects. (ii) *Semantic-confidence suppression*: for multiple similar instances inside O_1 , diffusion features carry spatial positional information, so cross-instance matches yield low or negative semantic cosine $\hat{s}_j$; the quadratic weight $w_j=\text{ReLU}(\hat{s}_j)^2$ drives these ambiguous point drifts to zero. This is the same mechanism validated by the similarity-skewness filter (Section 3.1) and Ablation 1 (Section 4.6), where any smooth non-linear weighting suppresses unaligned clutter.

E ON TECHNICAL NOVELTY

While Soft-VSM builds on diffusion-feature extraction and the dual-branch aggregation of Eldeskey et al. (2025), its contributions target metric formulation and stability rather than backbone design. (i) *Formulation*: both scalar thresholds are removed in closed form via non-linear confidence weighting and continuous visual integration ($\int_0^1 \mathbb{I}[\text{clamp}(\hat{v}_j, 0, 1) > r] dr = \text{clamp}(\hat{v}_j, 0, 1)$), eliminating step-discontinuities and point dropouts. (ii) *Robustness*: Ablation 6 shows hard VSM swings by up to 0.05 Pearson across visual-loss weights α , whereas Soft-VSM is virtually invariant ($r \in [0.635, 0.647]$). (iii) *Training*: an extended retraining protocol lifts oracle correlation from 0.467/0.558 to 0.638/0.674. (iv) *Differentiability*: Soft-VSM is the first fully differentiable dense visual-semantic consistency metric, usable directly as a loss or RL reward for diffusion personalization.